

SUPERSTORE SQL ANALYSIS

Data Analyst Portfolio Project Report

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Project: Superstore Sales & Profit Analysis

Database: PostgreSQL

Tool: pgAdmin 4

Query Language: SQL

Visualization: Power BI

Version Control: GitHub

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1. EXECUTIVE SUMMARY

The **Superstore SQL Analysis** project is a practical Data Analyst portfolio project designed to analyze retail and e-commerce transaction data using PostgreSQL and SQL.

The dataset contains **21,520 records** covering customers, products, sales, quantities, discounts, shipping methods, regions, and profits.

The project follows an end-to-end analytical workflow:

Data Import → Data Verification → Data Quality Assessment → KPI Analysis → Business Analysis → Insights → Recommendations

The main purpose of the project is to understand business performance and identify important patterns in sales and profitability.

The major KPIs calculated were:

KPI	Result
Total Sales	2,297,201.07
Total Profit	286,397.79
Total Orders	5,009

Profit Margin 12.47%

The analysis also covers regional performance, product performance, customer analysis, category analysis, shipping methods, and discounts.

The project demonstrates practical skills in SQL, PostgreSQL, data validation, KPI analysis, business analysis, data interpretation, and GitHub documentation.

2. INTRODUCTION

Data analysis helps businesses convert raw transaction data into useful information for decision-making.

Retail companies generate large amounts of transactional data every day. This data may contain information about customers, products, sales, discounts, shipping methods, and profit.

However, raw data does not directly explain business performance.

This project uses SQL to transform transaction-level data into meaningful business information.

The analysis focuses on identifying:

- Sales performance
 - Profitability
 - Order volume
 - Regional performance
 - Category performance
 - Product performance
 - Customer performance
 - Shipping performance
 - Discount impact
-

3. BUSINESS PROBLEM

The business needs to understand how its retail operations are performing.

Important questions include:

- What is the total revenue?
- How much profit is generated?
- What is the overall profit margin?
- Which regions perform best?
- Which categories generate strong sales?
- Which products generate high profit?
- Which products generate losses?
- Which shipping method is used most?
- How do discounts affect profitability?
- Which customers or segments contribute significantly?

Without proper analysis, it is difficult for management to identify areas of strength and areas requiring improvement.

SQL provides a structured method for answering these questions.

4. PROJECT OBJECTIVES

The objectives of this project are:

1. Import the Superstore dataset into PostgreSQL.
 2. Verify the imported data.
 3. Perform data-quality checks.
 4. Identify missing values.
 5. Identify duplicate records.
 6. Validate numerical fields.
 7. Validate date fields.
 8. Calculate important business KPIs.
 9. Analyze regional performance.
 10. Analyze categories and sub-categories.
 11. Analyze product performance.
 12. Analyze customer performance.
 13. Analyze shipping methods.
 14. Analyze discounts and profitability.
 15. Generate business insights.
 16. Provide business recommendations.
 17. Document the project on GitHub.
 18. Extend the analysis using Power BI.
-

5. DATASET INFORMATION

The project uses the **Superstore dataset**.

Dataset Details

Attribute	Details
Dataset	Superstore
Records	21,520
Database	ecommerce_analys is
Schema	public
Table	superstore
Database System	PostgreSQL
Interface	pgAdmin 4

Date Range

Order Date: 2014-01-03 to 2017-12-30

Ship Date: 2014-01-07 to 2018-01-05

6. DATASET COLUMNS

The table contains 21 columns:

1. row_id

2. order_id
3. order_date
4. ship_date
5. ship_mode
6. customer_id
7. customer_name
8. segment
9. country
10. city
11. state
12. postal_code
13. region
14. product_id
15. category
16. sub_category
17. product_name
18. sales
19. quantity
20. discount
21. profit

These fields provide information about transactions, customers, products, locations, shipping, sales, discounts, and profitability.

7. TECHNOLOGY STACK

PostgreSQL

PostgreSQL was used as the primary database for storing and analyzing the dataset.

pgAdmin 4

pgAdmin 4 was used to:

- Create the database
- Import the dataset
- Execute SQL queries
- View query results
- Perform data validation

SQL

SQL was used for:

- Data verification
- Data-quality checks
- Aggregation
- Filtering
- KPI calculation
- Business analysis

Power BI

Power BI is used to extend the SQL analysis into an interactive dashboard.

GitHub

GitHub is used to store and document the project.

8. DATABASE SETUP

A PostgreSQL database was created with the name:

ecommerce_analysis

The Superstore dataset was imported into:

public.superstore

The database structure was verified before starting the analysis.

9. DATA IMPORT

The Superstore CSV file was imported into PostgreSQL using pgAdmin 4.

During the import process, several practical CSV configuration issues were handled, including:

- Header row configuration
- Date formatting
- CSV quote settings
- Escape settings
- Column mapping

After import, sample records were checked to verify successful loading.

Example:

```
SELECT *
```

```
FROM superstore
```

```
LIMIT 10;
```

10. DATA VERIFICATION

Data verification was performed before analysis.

The following were checked:

- Table structure
- Column names
- Sample records
- Record count
- Date fields
- Numerical fields
- Missing values
- Duplicate records

This step is important because incorrect data can lead to incorrect business conclusions.

11. DATA QUALITY ASSESSMENT

Data quality checks were performed before calculating business KPIs.

NULL Value Check

The completed NULL check returned:

0 NULL values

This indicates that no missing values were identified in the checked dataset.

Why This Matters

Missing values can affect:

- Calculations
- Aggregations
- Charts
- Business conclusions

Therefore, checking NULL values is an important part of the Data Analyst workflow.

12. DUPLICATE CHECK

A duplicate check was performed on the dataset.

Result

No duplicate records were found.

This provides additional confidence that the dataset can be used for further analysis.

13. NUMERICAL DATA VALIDATION

Important numerical fields were checked.

Sales

Minimum:

0.44

Maximum:

22,638.48

Quantity

Minimum:

1

Maximum:

14

Discount

Minimum:

0.00

Maximum:

0.80

Profit

Minimum:

-6,599.98

Maximum:

8,399.98

The negative profit value is important because it shows that some transactions generated losses.

14. DATE VALIDATION

The date fields were checked before performing time-based analysis.

Order Date

2014-01-03 to 2017-12-30

Ship Date

2014-01-07 to 2018-01-05

Date validation helps ensure that trend analysis and shipping analysis are based on valid date information.

15. KPI ANALYSIS

Key Performance Indicators were calculated to understand overall business performance.

KPI	Value
Total Sales	2,297,201.07
Total Profit	286,397.79
Total Orders	5,009
Profit Margin	12.47%

Profit Margin Calculation

Profit Margin =

$(\text{Total Profit} / \text{Total Sales}) \times 100$

$(286,397.79 / 2,297,201.07) \times 100$

= 12.47%

The profit margin provides a useful baseline for comparing different business dimensions.

16. REGIONAL ANALYSIS

Regional analysis was performed to compare business performance across regions.

The analysis considered:

- Order count
- Sales
- Profit
- Profit margin

Example SQL:

```
SELECT  
  
    region,  
  
    COUNT(DISTINCT order_id) AS order_count,  
  
    SUM(sales) AS total_sales,  
  
    SUM(profit) AS total_profit,  
  
    ROUND((SUM(profit) / SUM(sales)) * 100, 2) AS profit_margin  
  
FROM superstore  
  
GROUP BY region  
  
ORDER BY total_sales DESC;
```

Business Purpose

Regional analysis helps identify:

- Strong markets
 - Lower-performing markets
 - Regional profitability
 - Potential business opportunities
-

17. CATEGORY ANALYSIS

Category analysis compares the performance of major product categories.

The analysis focuses on:

- Sales
- Profit
- Profit margin
- Order volume

This helps identify which product categories contribute most to the business.

Sales should be evaluated together with profit because a high-sales category may not necessarily be the most profitable.

18. SUB-CATEGORY ANALYSIS

Sub-category analysis provides a more detailed view of product performance.

It helps identify:

- High-sales sub-categories
- High-profit sub-categories
- Low-profit sub-categories
- Loss-making areas

This information can support product and pricing decisions.

19. PRODUCT ANALYSIS

Product-level analysis was performed to identify high-performing and low-performing products.

The analysis considers:

- Product name
- Sales
- Profit
- Profit margin

Product analysis is important because individual products can have significantly different profitability.

20. TOP PERFORMING PRODUCT

One of the strongest products identified was:

Canon imageCLASS 2200 Advanced Copier

Metric	Value
--------	-------

Sales	61,599.8 3
-------	---------------

Profit	25,199.9 4
--------	---------------

Profit Margin	40.91%
---------------	--------

This product generated substantial sales and a strong profit margin.

It is therefore an example of a high-value and profitable product.

21. LOW-PROFIT PRODUCT ANALYSIS

The project also examines products with low or negative profit.

This is important because high sales do not always mean high profitability.

Possible reasons for low profitability include:

- High discounts
- Low selling prices
- High product costs
- Low margins
- Operational expenses

Therefore, management should evaluate both revenue and profitability when making product decisions.

22. CUSTOMER ANALYSIS

Customer analysis helps understand how customers contribute to overall business performance.

Important measures include:

- Customer sales

- Customer profit
- Number of orders
- Customer segment
- Average order value

This analysis can help identify high-value customers and understand purchasing patterns.

23. CUSTOMER SEGMENT ANALYSIS

The dataset includes customer segments.

Segment analysis can compare:

- Sales
- Profit
- Orders
- Profit margin

This allows management to understand which customer groups contribute strongly to business performance.

24. SHIPPING ANALYSIS

Shipping performance was analyzed using the `ship_mode` column.

Completed Results

Ship Mode	Orders	Sales	Profit	Avg Sales/Order
Standard Class	2,994	1,358,216.08	164,089.45	227.58
Second Class	964	459,193.44	57,446.49	236.09

First Class	787	351,428.43	48,969.95	228.50
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Same Day	264	128,363.12	15,891.90	236.40
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Key Observation

Standard Class has the highest order volume, sales, and total profit among the analyzed shipping modes.

25. DISCOUNT ANALYSIS

Discount analysis examines the relationship between discount levels, sales, and profit.

Discounts can help increase sales, but excessive discounts can reduce profitability.

Therefore, the analysis should consider:

Sales + Profit + Discount

rather than sales alone.

This analysis can support better pricing and promotional decisions.

26. BUSINESS QUESTIONS

The project answers the following business questions:

1. What are the total sales?
2. What is the total profit?
3. How many distinct orders are present?
4. What is the overall profit margin?
5. Which regions perform strongly?
6. Which categories perform well?
7. Which sub-categories perform well?
8. Which products generate high sales?
9. Which products generate high profit?
10. Which products have poor profitability?
11. How do customers contribute to business performance?

12. Which customer segments perform well?
 13. Which shipping mode is most commonly used?
 14. How do shipping modes compare?
 15. How do discounts affect profitability?
-

27. KEY BUSINESS INSIGHTS

Insight 1 — Overall Business Performance

Total sales were:

2,297,201.07

Total profit was:

286,397.79

Overall profit margin:

12.47%

Insight 2 — Order Volume

The dataset contains:

5,009 distinct orders.

This represents the overall order activity analyzed in the project.

Insight 3 — Product Performance

The Canon imageCLASS 2200 Advanced Copier generated:

61,599.83 sales

and

25,199.94 profit

with a:

40.91% profit margin.

Insight 4 — Shipping

Standard Class was the dominant shipping method based on order volume, sales, and total profit.

Insight 5 — Profitability

Some transactions/products have negative profit.

Therefore, sales alone should not be used to measure business success.

Insight 6 — Discounts

Discount levels should be evaluated against profitability to avoid sacrificing profit for additional sales.

28. BUSINESS IMPACT

The analysis can support business decision-making in several areas.

Product Decisions

Identify profitable products and products that require review.

Pricing Decisions

Understand whether discounts may be reducing profitability.

Regional Decisions

Compare performance between different regions.

Customer Decisions

Identify valuable customer groups and customers.

Shipping Decisions

Understand the contribution of different shipping methods.

Management Reporting

Use Power BI to provide an interactive view of important KPIs.

29. BUSINESS RECOMMENDATIONS

Recommendation 1 — Focus on Profitable Products

Products with strong sales and healthy profit margins should receive greater attention.

Recommendation 2 — Review Loss-Making Products

Products generating negative profit should be reviewed for pricing, discounts, and costs.

Recommendation 3 — Improve Discount Strategy

Discounts should be monitored carefully to ensure that increased sales do not result in unnecessary profit loss.

Recommendation 4 — Monitor Regional Performance

Regions with lower profitability should be investigated to understand the underlying causes.

Recommendation 5 — Analyze Customer Value

High-value customers and customer segments can be studied further for retention and targeted business strategies.

Recommendation 6 — Monitor Shipping

Shipping methods should be evaluated based on order volume, profitability, customer expectations, and operational efficiency.

Recommendation 7 — Use Interactive Dashboards

Power BI can provide management with an easy-to-understand visual summary of business performance.

30. SQL CONCEPTS USED

The project demonstrates:

- SELECT
 - WHERE
 - GROUP BY
 - ORDER BY
 - HAVING
 - SUM
 - COUNT
 - COUNT(DISTINCT)
 - AVG
 - MIN
 - MAX
 - ROUND
 - CASE
 - Date functions
 - Date filtering
 - Aggregate functions
 - Subqueries
 - Joins
 - KPI calculations
 - Profit-margin calculations
 - Data-quality checks
 - Duplicate checks
-

31. DATA ANALYST SKILLS DEMONSTRATED

Technical Skills

- SQL
- PostgreSQL
- pgAdmin 4
- Database management
- Data validation
- Data cleaning
- Data aggregation
- Power BI
- GitHub

Analytical Skills

- Data exploration
 - Business problem solving
 - KPI development
 - Profitability analysis
 - Product analysis
 - Regional analysis
 - Customer analysis
 - Shipping analysis
 - Business insight generation
 - Data-driven recommendations
-

32. PROJECT WORKFLOW

Superstore Dataset



PostgreSQL Database



CSV Import



Data Verification



Data Quality Check



NULL Check



Duplicate Check



Numerical Validation



Date Validation



KPI Analysis



Regional Analysis



Category Analysis



Product Analysis



Customer Analysis



Shipping Analysis



Discount Analysis



Business Insights



Business Recommendations



GitHub Documentation



Power BI Dashboard



33. POWER BI DASHBOARD

The SQL analysis can be extended into an interactive Power BI dashboard.

KPI Cards

- Total Sales
- Total Profit
- Total Orders
- Profit Margin

Recommended Visuals

- Sales by Region
- Profit by Region
- Sales by Category
- Profit by Category
- Monthly Sales Trend
- Monthly Profit Trend
- Top 10 Products
- Shipping Mode Analysis
- Customer Segment Analysis

Recommended Slicers

- Year
- Region
- Category
- Segment
- Ship Mode

A dedicated Date table can be used to provide clean Year and Month filtering.

34. GITHUB PROJECT STRUCTURE

The project is organized as:

Superstore-SQL-Analysis/

|

|— README.md

|

```
|— superstore_analysis.sql
|
|— data/
|   └─ superstore.csv
|
|— screenshots/
|   └─ dashboard_overview.png
|   └─ regional_analysis.png
|   └─ category_product_analysis.png
|   └─ trend_shipping_analysis.png
|
|— dashboard/
|   └─ superstore_dashboard.pbix
|
└─ report/
    └─ Superstore_SQL_Analysis_Project_Report.pdf
```

35. PROJECT SCREENSHOTS

Four dashboard screenshots are sufficient for the portfolio.

Screenshot 1 — Dashboard Overview

Shows:

- Total Sales
- Total Profit
- Total Orders
- Profit Margin
- Main charts

- Dashboard filters

Filename:

dashboard_overview.png

Screenshot 2 — Regional Analysis

Shows:

- Regional sales
- Regional profit
- Regional comparison

Filename:

regional_analysis.png

Screenshot 3 — Category & Product Analysis

Shows:

- Category performance
- Sub-category performance
- Top products

Filename:

category_product_analysis.png

Screenshot 4 — Trend & Shipping Analysis

Shows:

- Monthly trends
- Shipping performance
- Related KPIs

Filename:

trend_shipping_analysis.png

36. FUTURE IMPROVEMENTS

Future improvements include:

1. Complete the Power BI dashboard.
 2. Add advanced SQL window functions.
 3. Add monthly and yearly trend analysis.
 4. Perform customer retention analysis.
 5. Perform customer lifetime-value analysis.
 6. Perform deeper discount-profit analysis.
 7. Add product ranking analysis.
 8. Add automated dashboard refresh.
 9. Add additional business KPIs.
 10. Improve interactive dashboard design.
-

37. PROJECT CHALLENGES

During the project, practical database and data-import challenges were encountered.

These included:

- PostgreSQL configuration
- CSV import settings
- Date formatting
- CSV quote/escape settings
- Database access/password issues
- Data validation

Solving these issues provided practical experience in working with a real database environment rather than only writing isolated SQL queries.

38. FINAL CONCLUSION

The **Superstore SQL Analysis** project demonstrates a complete Data Analyst workflow using PostgreSQL and SQL.

The project began with raw transaction data and progressed through:

Data Import → Data Validation → Data Quality → KPI Analysis → Business Analysis → Insights → Recommendations

The analysis identified:

- **Total Sales:** 2,297,201.07

- **Total Profit:** 286,397.79
- **Total Orders:** 5,009
- **Profit Margin:** 12.47%

The project also analyzed products, customers, categories, regions, shipping methods, and discounts.

The project demonstrates practical SQL knowledge, database handling, data-quality assessment, analytical thinking, business understanding, and data-driven recommendations.

It is suitable as a **Data Analyst fresher portfolio project** and provides practical evidence of SQL and analytical skills.

39. PORTFOLIO DESCRIPTION

Superstore SQL Analysis | PostgreSQL

Analyzed **21,520 Superstore transactions** using PostgreSQL and SQL to evaluate sales, profit, orders, profit margin, regional performance, products, customers, shipping methods, and discounts. Performed data-quality checks, calculated business KPIs, identified performance patterns, and translated SQL results into business insights and recommendations. The project was documented on GitHub and extended toward an interactive Power BI dashboard.

40. RESUME PROJECT DESCRIPTION

Superstore Sales & Profit Analysis | PostgreSQL, SQL, Power BI

- Analyzed **21,520 retail transactions** using PostgreSQL and SQL.
 - Performed data-quality checks, duplicate validation, KPI calculations, and business analysis.
 - Analyzed regional, category, product, customer, shipping, and discount performance.
 - Identified profitability patterns and generated data-driven business recommendations.
 - Documented the project on GitHub and extended the analysis toward a Power BI dashboard.
-

41. INTERVIEW EXPLANATION

"Tell me about your Superstore project."

I worked on a Superstore Sales and Profit Analysis project using PostgreSQL and SQL.

The dataset contained 21,520 transaction records. I first imported the data into PostgreSQL and performed data verification and quality checks, including NULL and duplicate checks.

After validating the data, I calculated important KPIs such as total sales, total profit, total orders, and profit margin.

Then I analyzed regional performance, categories, products, customers, shipping methods, and discounts.

One important finding was that the Canon imageCLASS 2200 Advanced Copier generated 61,599.83 in sales and 25,199.94 in profit, with a 40.91% profit margin.

I also analyzed shipping performance and found that Standard Class had the highest order volume, sales, and total profit.

Finally, I converted the analysis into business insights and recommendations and documented the project on GitHub. I am also extending the project into a Power BI dashboard.

42. PROJECT COMPLETION CHECKLIST

Database

- ✓ PostgreSQL database created
- ✓ `ecommerce_analysis` database created
- ✓ `public.superstore` table created
- ✓ Dataset imported
- ✓ Table verified

Data Quality

- ✓ NULL check completed
- ✓ Duplicate check completed
- ✓ Numerical validation completed
- ✓ Date validation completed

Analysis

- ✓ KPI analysis completed
- ✓ Regional analysis completed
- ✓ Category analysis completed

- ✓ Product analysis completed
- ✓ Shipping analysis completed
- ✓ Business insights prepared
- ✓ Business recommendations prepared

GitHub

- ✓ GitHub account created
- ✓ Superstore-SQL-Analysis repository created
- ✓ Repository made public
- ✓ README.md created
- ✓ README.md updated
- ✓ superstore_analysis.sql uploaded

Remaining

- ☐ Complete Power BI dashboard
 - ☐ Capture 4 dashboard screenshots
 - ☐ Upload screenshots
 - ☐ Upload Power BI file
 - ☐ Add final PDF report
-

43. PROJECT SUMMARY

Project Component	Details
Project	Superstore SQL Analysis
Role	Data Analyst Project
Dataset	Superstore
Records	21,520

Database	PostgreSQL
Tool	pgAdmin 4
Language	SQL
Total Sales	2,297,201.07
Total Profit	286,397.79
Orders	5,009
Profit Margin	12.47%
Visualization	Power BI
Version Control	GitHub
Repository	Superstore-SQL-Analysis

END OF REPORT

Superstore SQL Analysis
Data Analyst Portfolio Project
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